

Bearing Life Prediction Based on Vibration Signals

A Case Study and Lessons Learned

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Abstract—This paper presents the Professional-category winning algorithm of bearing Remaining Useful Life (RUL) prediction for the 2012 IEEE PHM challenge problem. The algorithm consists of extraction of bearing characteristic frequency features with envelop analysis, fault detection with PCA, and two RUL prediction strategies to address the scenarios when the bearing faults have and have not been detected. The paper will go through various aspects to investigate the challenge problem, synthesize modeling strategies, and summarize the lessons learned from this bearing life prediction case study.

Keywords—bearing; envelop analysis; prognostics; remaining useful life; principal component analysis; vibration

I. INTRODUCTION

Bearings are the most common components in rotary machines and usually constitute a large portion of the failure cases in these machines. Although there has been a long history in the study of bearing failure mechanisms, bearing life modeling and the development of bearing fault detection and prognostics techniques, the problem is still far from being solved. Bearing Remaining Useful Life (RUL) prediction is still a challenging task in academia and industries, especially for those approaches based on vibration signals. On the one hand, bearing vibration signals can reveal very specific information about the bearing's fault conditions, which is not detectable as early, if possible, through other means. On the other hand, vibration signals are subject to noise and also subject to variations from one system to another, which creates the challenge for signal processing and the building of automatic monitoring program that can adapt to each instance of the system. In addition, progression of vibration based features is not necessarily monotonic during the degradation of the bearing, which makes RUL prediction using vibration based features even more challenging.

Many research efforts on bearing RUL prediction have been reported every year, each claiming the effectiveness of the proposed algorithm based on the experimental results on a certain data set. However it is hard to compare those methods due to the different assumptions they make and the variability in data quality, quantity, and experimental settings. Admittedly, public, high-quality, realistic bearing run-to-failure data sets are very rare. Therefore the 2012 IEEE PHM conference brought to public a new bearing data set with a defined challenge

problem of bearing RUL prediction [1]. This paper presents the winning algorithm for the challenge problem in the Professional/Industry category and discusses lessons learned during the process of solving the problem.

The rest of the paper is organized as follows. Section II describes the challenge data set and the objective. Section III describes the analytical methods and some insights to the data that lead to a solution to the problem. Section IV discusses the prediction results and performance. Section VI concludes the study.

II. CHALLENGE DATA SET AND OBJECTIVE

The challenge data set includes run-to-failure vibration and temperature data collected from 17 bearings of the same model with Accelerated Life Tests (ALT). These ALTs are conducted under three operating conditions with combinations of different loads and rotational speeds as listed in Table I. The bearing model under test is a ball bearing whose characteristic parameters are listed in Table II. Out of the 17 bearing cases, 6 are provided with complete run-to-failure data, called the learning set, which is used to develop models. The rest 11 cases are provided with censored life data from new up to a certain time before failure, called the test set. The numbers of cases that belong to the three operating conditions and the two subsets respectively are listed in Table III.

The vibration data for each case include two channels from two accelerometers placed radially on the external race of the bearing in vertical (channel 1) and horizontal (channel 2) directions respectively. The load is applied to the bearing radially in horizontal direction. The vibration data are collected every 10 seconds for a period of 1/10 second at a sampling frequency of 25.6 kHz, which means 2560 samples per 10 second per channel. The temperature data include a single channel collected continuously at a sampling frequency of 10 Hz. Temperature data are not available for all cases though.

Failure of bearing is claimed when the amplitude of the vibration signal exceeds the threshold of 20 g (this is a vague statement from the competition description. See section III.A for clarification), after which the test is stopped to prevent propagation of damage to the whole system. The total life of each case in the learning set and the censored life of that in the test set are listed in Table IV.

TABLE I. BEARING OPERATING CONDITIONS

Symbol	Description	Cond. 1	Cond. 2	Cond. 3
L	Load (N)	4,000	4,200	5,000
N	Speed (rev/min)	1,800	1,650	1,500

TABLE II. BEARING CHARACTERISTIC PARAMETERS

Symbol	Description	Value
L_s	Load ratings static (N)	2,470
C	Load ratings dynamic (N)	4,000
N_{max}	Maximum speed (rev/min)	13,000
D_b	Diameter of rolling elements (mm)	3.5
n_e	Number of rolling elements	13
D_i	Diameter of inner race (mm)	22.1
D_o	Diameter of the outer race (mm)	29.1
D_p	Bearing mean diameter (mm)	25.6
ϕ	Contact angle (rad)	0

TABLE III. NUMBER OF BEARING CASES IN EACH CATEGORY

Category	Cond. 1	Cond. 2	Cond. 3
Learning set	2	2	2
Test set	5	5	1

TABLE IV. TOTAL LIFE/CENSORED LIFE (IN SECOND) OF EACH BEARING

Category	Index	Cond. 1	Cond. 2	Cond. 3
Learning set	1	28,020	9,100	5,140
	2	8,700	7,960	16,360
Test set	3	18,010	12,010	3,510
	4	11,380	6,110	-
	5	23,010	20,010	-
	6	23,010	5,710	-
	7	15,010	1,710	-

Each of the 17 cases is assigned with a unique identifier in the form of *Bearing_i_j*, with *i* denoting the index of the operating conditions and *j* denoting the case index. Those cases with *j*=1 or 2 belong to the learning set and *j*>2 the test set.

The objective of the challenge is to predict the RUL for each bearing case in the test set. The challenge comes with the following metric to evaluate submitted RUL predictions by comparing them with the RUL ground truth that was not released until competition ended. Let R_k ($k=1, \dots, 11$) be the RUL ground truth of the 11 cases in the test set and $\hat{R}_k$ be the corresponding RUL prediction. Firstly, the percentage error of an RUL prediction is calculated:

$$\%Er_k = 100 \times \frac{R_k - \hat{R}_k}{R_k} \quad (1)$$

Then the score of accuracy of each RUL prediction is defined using asymmetric penalties to late and early predictions, with late predictions penalized more:

$$S_k = \begin{cases} \exp(-\ln(0.5) \times Er_k / 5) & \text{if } Er_k \leq 0 \\ \exp(+\ln(0.5) \times Er_k / 20) & \text{if } Er_k > 0 \end{cases} \quad (2)$$

The final score of a submission with all 11 predictions is then defined as the average of all:

$$\text{Score} = \frac{1}{11} \sum_{k=1}^{11} S_k \quad (3)$$

III. ANALYSIS AND MODELING

The process of RUL modeling follows the general procedures of feature extraction, health assessment (to compute health index), fault detection, and life prediction (either through health index extrapolation with failure criteria evaluation, or through direct inference from health index to RUL). However, one more thing to do before all these procedures is to clarify the failure criteria because it may have big influence on the choice of detailed modeling procedures.

A. Clarification of Failure Criteria

It has not been explicitly stated in the competition description which one of the two vibration signals are used to define failure. Through investigation of both vibration signals for the entire life of each case, it is found that the vibration measurement in horizontal direction (load direction) may exceed the failure threshold long before the test ends while the other measurement only exceeds the failure threshold toward the end of life in the learning set. Therefore the signal for vertical direction should be the one that defines failure. In addition, we can clarify that it is the absolute amplitude of the raw signal (not RMS value or other computed value) that is used to evaluate against the failure threshold.

If the above failure definition is taken for RUL modeling, the RUL prediction problem will then be defined as the prediction of the First Passage Time of the raw vibration signal. This formulation is problematic because it forces the prediction model, no matter what procedures it takes, to make predictions to the absolute value of the raw vibration signal in time domain, which is an inappropriate approach based on common sense. Then we have to find some other failure criteria that can be translated to the existing one but allows direct evaluation in the domain where the predictions are made. Or alternatively, we can even find some criteria that are directly defined using the end-of-life signature of the bearings, disregarding the original quantitative failure threshold. For example, if a prediction model is based on predictions on the RMS of the raw signal, rather than trying to find the relations between RMS and the absolute amplitude of the signal, we can directly establish statistical failure criteria using RMS values near the end life of all the learning cases. This paper will continue with the second approach, i.e. to disregard the existing amplitude-based failure criteria and derive other criteria, if needed, directly from the run-to-failure data.

B. Feature Extraction

Since the temperature signal is not consistently available for all learning and test cases, only the vibration signal is considered in the rest of the paper. Features are extracted from each of the 0.1-second-long vibration measurements for the entire life of each case, including time domain and frequency domain features. Time domain features are as simple as max, RMS, and Kurtosis of the samples for each measurement.

Extraction of frequency domain features requires the knowledge of bearing degradation and failure characteristics.

Ball bearings usually have four failure modes, inner race fault, outer race fault, ball fault and case/train fault. Each failure mode can be explained by one characteristic frequency, whose value can be calculated analytically from the bearing geometric parameters (see Appendix A). These frequencies for the bearing model in this study are listed in Table V. They are multipliers of the fundamental frequency of shaft rotation f_N .

It is well known knowledge that the typical bearing degradation process can be divided into four stages along the time of usage (refer [2] for detailed explanation). In the first stage, defect signatures show up in the ultrasonic frequency ranges (>20 kHz), which is out of the reach of the existing data set due to sampling frequency limitations. In the second stage, the bearing characteristic frequencies are modulated by certain natural frequencies of the system (500 Hz~2000 Hz) and appear as sideband frequencies around the natural frequencies. Modulation means the characteristic frequencies are multiplied with the natural frequencies rather than added to them. As a result, energies at the characteristic frequencies may be suppressed to a much lower level while the majority of the energy appears as the sideband energy around the modulating frequency. Such problem can be addressed by envelop analysis [3-4], a popular technique for bearing fault analysis. When it comes to the third stage, the bearing characteristic frequencies can be directly observed in the frequency spectrum. In the fourth stage, the vibration amplitude increases greatly and all the harmonic amplitude reduces while noise dominates the frequency spectrum.

The time domain features can be useful when the bearing degrades to stage four. For stage two and three, frequency domain features are more useful. Details on the feature extraction methods will be explained below.

1) Frequency energy features

In order to extract representative features for the fundamental frequency and the bearing's characteristic frequencies, first apply FFT to each vibration measurement (2560 samples) to get the frequency spectrum. Define normalized energy at frequency f as

$$E_f = A_f^2 / \sum_{F>0} A_F^2 \quad (4)$$

where A_F is the amplitude of frequency F in the frequency spectrum. Note that the summation term in (4) excludes the energy for frequency $F=0$, which is the DC offset of the sensor. With this definition, normalized energies are computed at one (1x), two (2x) and three (3x) times of the fundamental frequencies as well as the four bearing characteristic frequencies given in Table V. Finally, the three normalized energies in each of the five categories sum up respectively to generate five features $E_N, E_{BPFO}, E_{BPFI}, E_{BFF}, E_{FTF}$.

2) Frequency energy features with envelop analysis

Envelop analysis includes two procedures applied to the original vibration signal (refer [4] for a tutorial):

TABLE V. BEARING DEFAULT FREQUENCIES

Symbol	Description	Calculation
f_{BPFO}	Ball pass frequency outer race	$5.6113 \times f_N$
f_{BPFI}	Ball pass frequency inner race	$7.3887 \times f_N$
f_{BFF}	Ball fault frequency (two times of the ball spin frequency)	$7.1776 \times f_N$
f_{FTF}	Fundamental train frequency	$0.4316 \times f_N$

- Apply a band pass filter to the signal. The band pass central frequency is chosen at a certain natural frequency of the system that is many times higher than the frequency of interest.
- Apply Hilbert transform to the filtered signal to obtain a new signal in time domain.

After these two steps, the regular frequency analysis methods such FFT can be applied to the envelop signal. In this study, the same set of features are extracted from the envelop signal following the procedures described in the previous section. By varying the envelop settings, i.e. the location of the band pass center and the selection of band pass width, multiple set of features can be generated. The selection of the envelop parameters here includes heuristic decisions based on the observations on the frequency spectrum of the signals, because the natural frequencies of the system is unknown and thus have to be estimated somehow (the formal approach to measure the system's natural frequencies is through a special "hammer test", which is not present in this study). In the frequency spectrum, a natural frequency of the system is the one that is invariant to the shaft fundamental frequency and has relatively high amplitude in nearby frequencies. An appropriate band width is chosen to cover the nearby high-amplitude frequency range. In this study, two band pass filters are designed for two potential natural frequencies, which correspond to two envelop settings as listed in Table VI.

3) Smoothing of feature time series

The extracted features form a multivariate time series. Due to the high noise level, the time series of each feature $\{s_t\}$ is smoothed by an exponential smoothing filter with a smoothing factor $\alpha=0.9$ to get the new time series $\{y_t\}$:

$$\begin{aligned} y_0 &= s_0 \\ y_t &= \alpha \cdot y_{t-1} + (1 - \alpha)s_t, t > 0 \end{aligned} \quad (5)$$

Fig. 1 shows an example of the smoothed features for Bearing2_1, some of which demonstrate clear trends towards the end life. By studying what features are trending up towards the end life, it is possible to roughly estimate the failure modes of the bearings in the learning set. For example, Bearing2_1 demonstrates a rising trend in the energies of BPFI and BFF towards the end life. Therefore this bearing may have failed due to inner race failure or ball failure. Need to mention that BPFI and BFF are very close to each other and thus are hard to distinguish given the frequency resolution of 10 Hz of the existing measurement (0.1 second per measurement). It is interesting that the energy of BPFO rises for a while but drops before the energies of BPFI and BFF start to rise rapidly toward the end life.

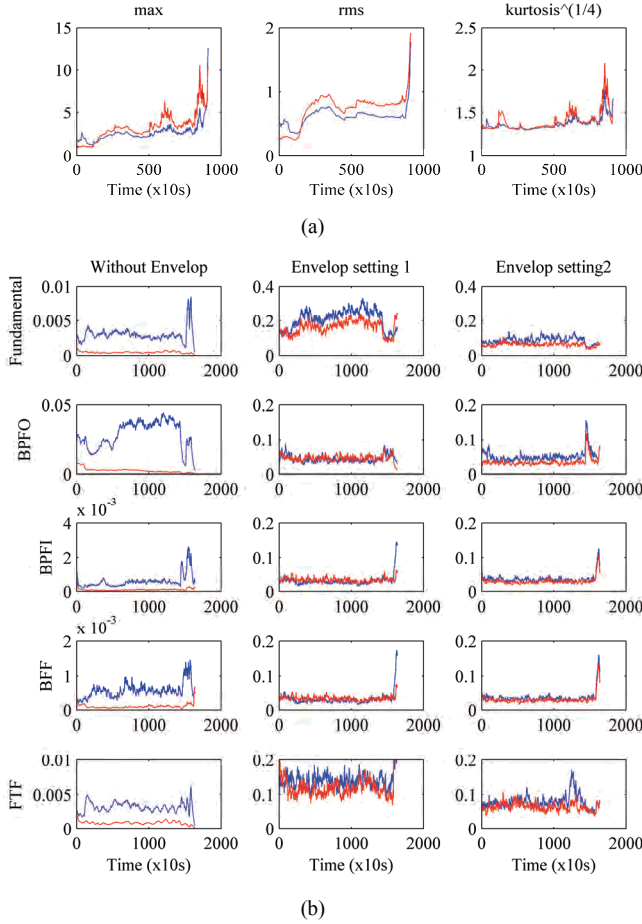


Figure 1. Smoothed time series of generated features for Bearing2_1. Each subplot shows two features from two channels of the vibration sensors (blue for vertical and red for horizontal respectively). The total number of features is 36. (a) time domain features (b) frequency domain features

The estimated failure modes of each bearing are listed in Table VII, even though failure mode estimation may or may not be necessary for bearing RUL prediction, which all depends on the particular prediction method employed.

C. RUL Prediction

Ideally, the RUL prediction problem can be formulated as a regression problem where an overall regression model between the extracted features and the corresponding RULs is learned using all the samples combined from 6 cases. This approach, however, is hardly practical in this case study due to the following reasons.

Firstly, most of the features with consistent degradation patterns (e.g. those energies at bearing characteristic frequencies calculated with envelop analysis) across all learning cases remain flat for over 90% of the life and only rise rapidly when the life is close to the end, which can be observed from the feature time series. A regression model trained on all the samples of the entire history may not be able to model the early life patterns well. On the contrary, a regression model can easily over fit the dynamic patterns that are inconsistently demonstrated by some other features.

TABLE VI. BAND PASS FILTER SETTINGS FOR ENVELOP ANALYSIS

Setting index	Center frequency (Hz)	Lower cut off frequency (Hz)	Higher cut off frequency (Hz)
1	1,100	600	1,600
2	3,500	2,500	4,500

TABLE VII. ESTIMATED BEARING FAILURE MODES

Case	Estimated failure modes
Bearing1_1	inner race, ball, cage
Bearing1_2	inner race, outer race, ball, cage
Bearing2_1	inner race, ball
Bearing2_2	outer race
Bearing3_1	outer race
Bearing3_2	inner race, ball, cage

Secondly, features from different bearing cases tend to have different noise level and offset. An overall regression model built upon all training cases may be difficult to generalize the case-to-case difference.

Therefore in this study the following RUL prediction strategy is used. Firstly, rather than looking for a regression model that can predict arbitrary long life, we limit the prediction horizon (i.e. how early ahead of an event a predictor can make predictions for) to the time point that a fault can be detected at the first place. Beyond such point, predictions will not be made from the condition data, but rather through certain reliability approaches that rely on life distributions and the operating conditions only. To achieve this, a fault detection model is needed to gate the follow-up RUL prediction methods. Secondly, each bearing case will have its own fault detection model built upon its own history. With this approach, there will be no need to model the operating conditions for different bearings.

1) Fault detection model

Because each case needs one model built on its own history, supervised fault detection models won't be applicable since a test case will not have known fault labels. In this study, the PCA (Principal Component Analysis) based fault detection approach is employed, which is a widely used anomaly detection method in statistical process control [5]. The so-called T^2 statistics are computed first. However, the statistical approach to set anomaly threshold in Hotelling's T^2 test won't be used here because the features (after time series smoothing) in this application may not meet the assumptions on data distributions. Therefore the T^2 statistics is only used as a health index while the anomaly threshold is determined through other means.

The training data consist of those samples in the early life (e.g. up to 100 measurement cycles before end life) of a bearing, noted as $X=(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)^T=(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_m)$, where $\mathbf{x}_n$ is a feature vector at time n , and $\mathbf{y}_m$ represents all the samples of the m^{th} feature. The test data consist of all samples of the case, noted as $X_T=(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_t)^T$. Then the fault detection process can be defined with the following procedures.

- Calculate mean and variance for each feature $\mathbf{y}_m$ in X .
- Normalize each feature $\mathbf{y}_m$ in X using its mean and variance to get normalized training data Z .

- c) Compute eigen decomposition for the covariance of Y : $Y^T Y = V \Lambda V^T$, where $V = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m)$ is the eigen vector matrix and $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$ the eigen value matrix. Here the eigen values in Λ are arranged in descending order such that $\lambda_1 \geq \lambda_2, \dots, \geq \lambda_m$.
- d) For a cutoff value p ($0 < p < 1$), find the first l eigen values such that

$$\sum_{i=1}^{l-1} \lambda_i < p \leq \sum_{i=1}^l \lambda_i$$

- e) Construct the sub matrix $V_{m \times l} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_l)$ and $\Lambda_{l \times l} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_l)$
- f) Normalize the test data X_T using the mean and variance calculated in step a) to get $Z_T = (\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_t)^T$.
- g) The T^2 statistics, or health index $h(t)$ can be computed for each samples in Z_T

$$h(t) = T^2(\mathbf{z}_t) = \mathbf{z}_t^T V_{m \times l} \Lambda_{l \times l} V_{m \times l}^T \mathbf{z}_t \quad (6)$$

- h) Claim fault at time t if $h(t)$ exceeds a certain threshold a .

The total number of features extracted is 36, including 3 statistical features, 5 frequency energy features, 10 frequency energy features with two envelop settings for two vibration sensors. In this study, only the 20 features extracted with envelop analysis are used in the aforementioned fault detection model because the other features exhibit highly dynamic and inconsistent behavior, which is compromising the accuracy of model. Of course using only the enveloped frequency features implies that we are only considering the situation of stage 2 degradation of the bearings, which is a decision customized for this case study only. The cut off value for selecting top eigen values in step d) is set to $p=0.95$. The threshold for fault determination in step h) is set to $a=50$ to balance false positive false negative based on computed health index time series of the 6 learning cases (Fig. 2-a).

2) RUL prediction with fault detected

Once the health index time series are prepared and the fault threshold selected, the time from fault detection to failure, called Detect-to-Fail in short, for the learning cases can be obtained, as listed in Table VIII. Detect-to-Fail reflects the prediction horizon that the condition based RUL prediction algorithm can make.

For the time period within the prediction horizon, it is possible to build a local regression model to predict RUL. The health index time series again demonstrates nonlinear, inconsistent progression behavior across different bearings, which makes the building of local RUL prediction models not a trivial task either. For this given test set, however, there's barely the need to do so since only one out of the 11 test cases was detected with fault. Therefore in this study, no local regression model is actually built to predict RUL within this relatively short period of time. Instead, the average Detect-to-Fail time under each condition is directly used to estimate the bearing's RUL.

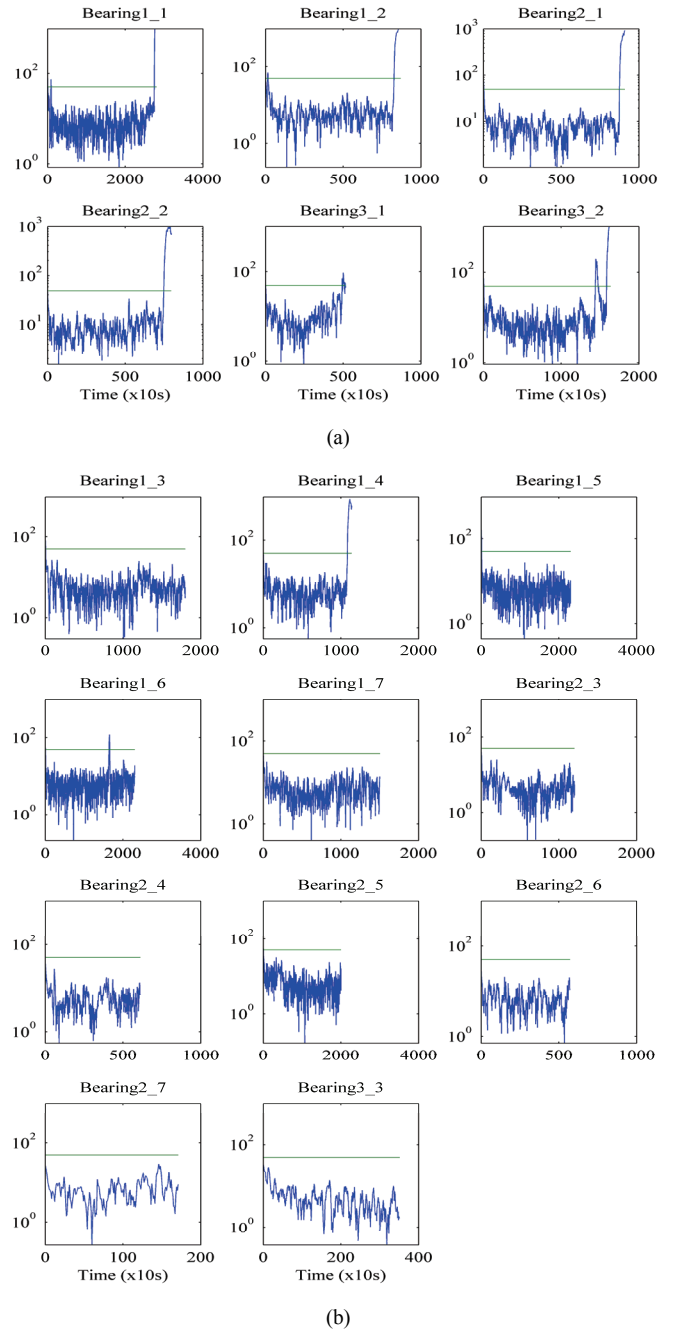


Figure 2. Bearing health indices in log scale vs. time. (a) Learning set. (b) test set.

For example, test case Bearing1_4 was detected to be faulty at 550 seconds before the censored end life. Since 550 seconds are already longer than the average Detect-to-Fail time of 490 seconds, we can expect the bearing to reach end life very soon. Therefore the RUL prediction for it is set to 1 measurement cycle or 10 seconds directly.

3) RUL prediction without fault detected

Before fault detected, almost no clear evidence of degradation can be obtained from the features extracted in this study. Therefore, prognostic models relying on condition monitoring are not functioning any more. Other approaches

such as reliability models can be used instead. Reliability models rely on the assumptions of bearing life distribution (usually Weibull). With the limited number of failure cases given in this study, however, it is difficult to estimate the parameters of such distributions accurately. Due to the limited time available for the challenge problem, the author did not continue with this approach. Instead, the average Detect-to-Fail time from the training cases are used again as the RUL predictions for each of the 10 test cases that were considered healthy, no matter how long the case has been running. This is a very conservative prediction. The bottom line is that this prediction provides a lead time that one can trust before a healthy bearing runs into failure.

IV. RESULTS AND DISCUSSIONS

The predicted RUL and the ground truth (revealed after the competition closed) are listed in Table IX for comparison. The total score computed using the evaluation metric defined in (1-3) is 0.0981, which is equivalent of 16.8% error for late predictions or 67.0% error of early predictions in average.

At the first glance of the comparison, we have to say that these predictions are by no means “good” ones from any perspectives because the error uncertainty is too big to yield any practical value. However, to directly infer the practical value from the context of a competition will not yield a fair conclusion. For sure, a competition needs to reduce the many design issues of a practical problem into single criteria in order to rank the participants. Therefore some real world design issues have to be sacrificed, for example:

- What prediction accuracy/uncertainty is *acceptable* for the application;
- What prediction horizon is *adequate* in an application;
- And what prediction horizon is *achievable* from algorithms based on the available data.

The performance criteria for a real-world RUL prediction problem can be much more complex than the single accuracy criteria as used in this paper. For example, a four-level performance evaluation framework was proposed in [6], which encompasses many aspects of a RUL prediction problem, including prediction horizon, accuracy, convergence, etc. In this case study, the prediction horizon of a vibration-based RUL prediction algorithm has been limited to the time that the bearing enters degradation stage 2.

Once reaching stage 2, the bearing’s RUL is usually 5~10% of the L_{10} life [2] before a complete failure at stage 4. The prediction horizon estimated in this study is relatively shorter than such an empirical estimate. One reason for the discrepancy is that with the failure definition in this study failure of bearing is claimed before it actually reaches stage 4.

In this case study, the author made very conservative RUL estimates when a case’s condition is beyond the prediction horizon. This can be improved with quantitative prediction if the prediction scoring function of the challenge is utilized to derive a risk model for stochastic predictions and combined with the bearing reliability models (if parameter estimation is possible).

TABLE VIII. TIME (IN SECOND) FROM FIRST FAULT DETECTION TO FAILURE OF THE BEARINGS IN THE LEARNING SET

Case Index	Cond. 1	Cond 2	Cond 3
1	540	350	220
2	440	470	1970 (500 ^a)

^a The first detected fault did not lead to failure directly in this case. The second one did.

TABLE IX. COMPARISON OF RUL PREDICTION AND THE GROUND TRUTH

Bearing ID	Current Life (sec)	RUL Ground Truth	RUL Prediction ^b (sec)
Bearing1_3	18,010	5,730	490
Bearing1_4	11,380	339	10
Bearing1_5	23,010	1,610	490
Bearing1_6	23,010	1,460	490
Bearing1_7	15,010	7,570	490
Bearing2_3	12,010	7,530	410
Bearing2_4	6,110	1,390	410
Bearing2_5	20,010	3,090	410
Bearing2_6	5,710	1,290	410
Bearing2_7	1,710	580	410
Bearing3_3	3,510	820	360

^b. The predictions have minor difference from those as the final submission in the challenge.

V. CONCLUSIONS AND REMARKS

This paper presents a bearing RUL prediction approach by going through the various steps to solve the PHM challenge problem. The final solution is composed of multiple widely used techniques such as envelop analysis and PCA. Some problems encountered during the development of such a solution may resonate with the common issues to many other prognostics applications.

Firstly, definition of failure is critical to build good models. We can imagine that the end-of-life degradation level of the 17 bearings in this study may not be necessarily similar, because the failure has been measured by the amplitude of the highly unstable vibration signal, which may introduce a lot of uncertainty to the life ground truth from the beginning. This issue is common in defining failures of most mechanical components and constitutes a major source of poor data quality.

Secondly, unlike fault detection and identification problems that have well-defined and widely-acceptable performance evaluation criteria, RUL prediction problems do not. No single criteria can well describe the real-world expectations for a prognostic algorithm. Therefore each problem may have its own performance criteria. Different performance criteria may favor different uncertainty management style (conservative or risky) of the prediction algorithms. Ideally, if the performance criteria can be well defined to reflect the value and impact to reality, it can be, and should be, used as the utility function to optimize and fine tune each algorithm.

Lastly, vibration based features as a time series tends to progress in a non-linear fashion who pattern varies from case to case. Such patterns are hard to generalize using regression models. Instead, it may be more appropriate to use discrete degradation stages to qualitatively describe the degradation pattern. Identifying the current degradation stage will directly give a rough RUL estimate and also allow the use of local prediction models for more accurate predictions.

APPENDIX

A. Characteristic frequency calculation for roller element bearings:

$$f_{FTF} = \frac{1}{2} \left(1 - \frac{D_b}{D_p} \cos(\varphi) \right) f_N$$

$$f_{BPFO} = \frac{n_e}{2} \left(1 - \frac{D_b}{D_p} \cos(\varphi) \right) f_N$$

$$f_{BPF1} = \frac{n_e}{2} \left(1 + \frac{D_b}{D_p} \cos(\varphi) \right) f_N$$

$$f_{BFF} = \frac{D_p}{D_b} \left(1 - \left(\frac{D_b}{D_p} \cos(\varphi) \right)^2 \right) f_N$$

n_e : Number of rolling elements

D_b : Diameter of rolling elements

D_p : Bearing mean diameter

φ : Contact angle

f_N : Shaft rotation frequency

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